

Write-up: Model Documentation

1. Who are you?

Name: sunday

Hometown: China

I am a freelancer and very interested in AI.

2. Motivation

I hope AI help human beings

3. High-Level Approach

Fine tuning with qwen3-asr-1.7b, ensemble with weight average(model soup)

4. Visualizations

None.

5. Three Most Impactful Code Segments

- use spec-aug to improve model generalize ability

```
if self.training and (self.mask_feature_prob > 0 or
self.mask_time_prob > 0):
    mixup_input_features =
self._mask_input_features(mixup_input_features,
attention_mask=mixup_feature_attention_mask)
```

- add noise to improve model generalize ability

```
trans.append(A.AddBackgroundNoise(sounds_path=["../data/processed/noise_16k"], min_snr_db=self.cfg.am_bn_noise_db,
max_snr_db=self.cfg.am_bn_noise_max_db,
p=self.cfg.am_bn_noise)) if self.cfg.am_bn_noise>0 else None
```

6. Machine Specs & Time

CPU (model): 13th Gen Intel(R) Core(TM) i7-13700KF

GPU (model or N/A): single RTX4090 24G

Memory (GB): 64G

OS: Ubuntu 22.04.2 LTS

Train duration: 144 Hours

Inference duration: ~118 minutes on datadrive env

7. Caveats & Known Issues

NA

8. Tools for Data Preparation / EDA

NA

9. Performance Evaluation Beyond Competition Metric

I only use the provide metric

10. Things Tried That Didn't Make Final

I use qwen3-tts to generate more training data, but the result is not good.

11. Future Improvements

- The training data is noisy, maybe we need do some data curation

12. Simplifications for Faster Inference

Simply use only 1 model without ensemble, use beam size 1 not 2.